DOF-Based Mechanism Design of RigFlex Parallel Robots

Xiao Wang[1], Shuo Yang[2], Zihe Lei[1], Ke Wu[3,†], Guohua Chen[1,*]

***Abstract*— Rigid-flexible (RigFlex) parallel robots integrate the advantages of both rigid and flexible robots, offering significant potential for practical applications. However, current research on RigFlex robots remains limited, particularly due to the lack of a systematic design methodology. To address this gap, this paper proposes a design methodology for RigFlex parallel robots based on the degree of freedom (DOFs) of rigid parallel robots. The design methodology proposed in this paper is applied to the development of RigFlex parallel robots using a 3-RRR parallel robot as an example. Initially, three RigFlex parallel robots were designed based on the 3-RRR parallel robot. Subsequently, a kinetostatics model was conducted on these designs, and numerical methods were employed to solve the corresponding models. Finally, the resulting motion and deformation characteristics of the designed RigFlex parallel robots were obtained, thereby validating the feasibility of the proposed design methodology. Furthermore, this study fills a research gap in the field of RigFlex robotics and provides a systematic framework to guide the design process of such robotic systems.**

I. INTRODUCTION

Rigid robots, owing to their high rigidity, strong load-bearing capacity, and precise control, have been widely adopted across various industries. However, as application scenarios diversify, the lack of compliance in rigid robots has become a significant challenge [1], [2]. In contrast, flexible robots, characterized by their compliance and deformability, ensure safety during interactions with the environment, making them suitable for applications where rigid robots are insufficient. Nevertheless, flexible robots are constrained by their lower rigidity and reduced control accuracy. RigFlex robots effectively combine the advantages of both rigid and flexible robots and have emerged as a prominent research focus in the field [3].

Rigid robots pose safety risks during interactions with the external environment. To address this issue, several researchers have conducted studies aimed at enhancing the compliance of rigid robots. Escarabajal et al. proposed a combined admittance control method using Dynamic Movement Primitives (DMPs) to ensure safe human-robot interaction during operation [4]. The academic work by Qi et al. addresses the challenge of controlling vertical jumps with centroidal angular momentum (CAM) and landing impact absorption for humanoid robots. They propose a method called CoP-Guided Angular Momentum Control and Impact Absorption [5]. Dehio et al. proposed an impedance control method to enable human-multi-robot collaboration scenarios [6]. Shi et al. proposed a real-time and guaranteed method for whole-body safe tracking control to ensure the safety of space robots in future complex autonomous on-orbit services [11]. Ba et al. investigated the impact of the dynamics characteristics of the leg hydraulic drive system on robot impedance control and proposed a dynamics compensation method for impedance-based motion control [8]. Tan et al. proposed an obstacle avoidance tracking control method to ensure the safety of automated lifting robots during their interaction with the surrounding environment [9].

Although flexible robots offer advantages due to their compliance, flexibility, and hyper-redundancy, they suffer from low control accuracy and reduced stiffness. To address these issues, several researchers have conducted studies aimed at enhancing the control precision of flexible robots. Huang et al. proposed a comprehensive modeling and control framework to address the issue of soft robots being unable to achieve precise and robust control under uncertain forces and contact with the environment [10]. Shi et al. employed an online iterative learning control method to ensure that soft robots achieve precise gait movements comparable to those of rigid robots [11]. Zhao et al. proposed a redundantly controlled algorithm to achieve vibration suppression in flexible robots, addressing the issue of low control accuracy in cable-driven parallel robots [12]. Jamwal et al. implemented a fuzzy-based closed-loop feedback controller for compliant parallel robots to improve their motion control accuracy [13]. Sozer et al. proposed a novel fluidic-driven variable stiffness revolute joint to address the controllability and predictability challenges inherent in soft robotics [14]. Yang et al. addressed the issue that passive compliance is insufficient to prevent soft grippers from crushing the target, and proposed an online learning self-tuning nonlinearity impedance controller to solve this problem [15]. Tang et al. proposed a learning-based optimal control approach to address the issue of ineffective control in soft robot-environment interactions [16].

Current research indicates that rigid robots, compared to flexible robots, offer higher control accuracy. However, they are prone to rigid impacts during interactions with the external environment, which poses safety risks. So most existing studies on rigid robots focus on enhancing their compliance. In contrast, flexible robots offer the advantage of compliance during interactions with the environment, unlike rigid robots. However, their control accuracy is generally

* denotes the corresponding author
† denotes the co-corresponding author
[1] Intelligent Robot and Unmanned Technology Laboratory, Beijing University of Chemical Technology, Beijing 100029, China. Corresponding author Guohua Chen, e-mail: chengh@mail.buct.edu.cn
[2]College of Mechanical and Electrical Engineering, Beijing University of Chemical Technology, Beijing 100029, China.
[3]Robotics department, Mohamed bin Zayed University of Artificial Intelligence, Masdar City, Abu Dhabi, United Arab Emirates.

lower. Research on flexible robots primarily aims to improve their control accuracy to match that of rigid robots. Designing a rigid-flexible robot that integrates the advantages of both rigid and flexible robots has become a significant research direction in the field of robotics. Current research on rigid-flexible robots primarily focuses on the following aspects. Jing et al. proposed a novel structure for a continuum manipulator that addresses the issue of low stiffness in pneumatic-drive soft manipulators. The new design integrates both rigid and flexible structures, resulting in a rigid-flexible coupling manipulator (RFCM) [17]. Liu et al. conducted an analysis of the natural frequency characteristics of spatial rigid-flexible coupled parallel mechanism systems [18]. Rong et al. applied the discrete-time transfer matrix method for multibody systems to achieve fast modeling and computation of the dynamics of a planar closed-chain rigid-flexible five-bar mechanism [19]. Cervantes-Culebro et al. proposed a concurrent optimal design for a planar five-bar parallel robot aimed at high-speed pick-and-place tasks [20].

Current research on the mechanism design of RigFlex parallel robots is limited, and no specific design methodology has been proposed to date. This paper proposes a design methodology for RigFlex parallel robots based on DOFs. In section II, a comprehensive design framework is presented, and in section III, the methodology is applied to a 3-RRR parallel robot as a case study. First, a RigFlex parallel robot is designed based on 3-RRR parallel robot. Subsequently, a kinetostatics model is developed and solved using numerical methods. The motion performance of the designed RigFlex parallel robot is then analyzed, thereby validating the feasibility of the proposed design methodology. The proposed design methodology addresses a critical gap in the field of RigFlex robotics, providing a systematic framework to guide the design process of such robotic systems. The contributions of this paper are summarized as follows

1) This study analyzes the DOFs in mechanisms, examines the relationship between mechanical constraints and actuation, and proposes a design methodology for RigFlex parallel robots based on these principles.
2) Using the 3-RRR parallel robot as an example, three types of RigFlex parallel robots were designed based on the methodology proposed in this paper.
3) Kinetostatics models were developed for the three designed RigFlex parallel robots, and the results were solved using numerical methods.

II. Overview of the proposed methodology

When the DOFs of a parallel robot are lower than the theoretical maximum available to a free rigid body in its motion space, the robot's motion becomes constrained. In other words, the robot is not 'completely free' but rather a constrained system [21]. For such constrained mechanisms, operating under redundant actuation conditions (i.e., the number of actuators exceeds the robot's actual DOFs), any excess actuation force that cannot be fully converted into useful motion will generate additional internal stresses, leading to elastic deformation of the components [22]. Under ideal conditions, if the mechanism is perfectly symmetric, the redundant actuation forces may cancel each other out, thereby preventing or minimizing deformation. However, for a zero-DOF (DOF = 0) robot or a negative-DOF robot, since the mechanism inherently lacks motion capability, all applied actuation forces are converted into internal stresses, inevitably causing deformation.

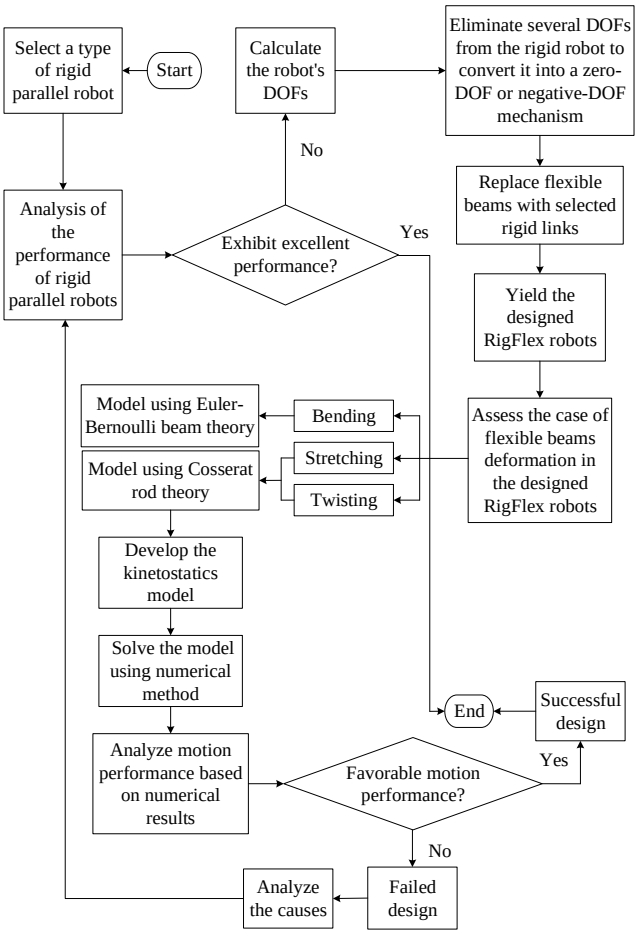


Fig. 1: Flowchart of the iterative design methodology.

As shown in the Fig.1, Based on the principles outlined above, this paper proposes a methodology for designing a RigFlex parallel robot based on the robot's DOFs. Initially, a rigid parallel robot is selected as the baseline model. Subsequently, a comprehensive performance analysis is conducted, focusing on critical parameters such as singularity avoidance capability and workspace size. If the analysis indicates satisfactory performance, further modifications to the robot's structure are deemed unnecessary. However, if the robot exhibits suboptimal performance, a RigFlex parallel robot is designed to enhance its functional characteristics. During the design process, the first step involves calculating the DOFs of the rigid parallel robot and determining its maximum attainable DOFs within the workspace. Following this, several DOFs are strategically eliminated to transform the robot into a zero-DOF or negative-DOF configuration. This transformation is implemented to ensure that the flexible

beams integrated within the designed mechanism undergo the desired deformation during operation, thereby achieving the intended improvements in performance. Subsequently, the deformation behavior of the flexible beams during motion is analyzed. If the flexible beams in the RigFlex robot exhibit bending deformations during operation, the geometrically nonlinear Euler–Bernoulli beam theory is employed for modeling [27]:

$$EI\frac{\mathrm{d}\theta}{\mathrm{d}s} = M(s) \tag{1}$$

Where E denotes the elastic modulus, I represents the second moment of area of the cross-section, M signifies the applied moment, and $\frac{\mathrm{d}\theta}{\mathrm{d}s}$ indicates the beam curvature. However, if they undergo axial stretching or twisting deformations, the Cosserat rod theory is utilized instead [27]:

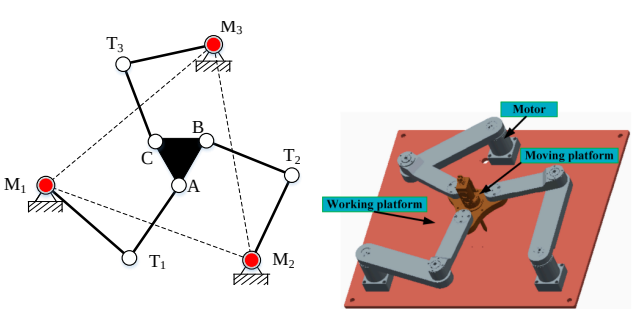
$$\begin{aligned} \frac{\partial \mathbf{r}(s)}{\partial s} &= \mathbf{v}(s), \qquad \frac{\partial \mathbf{R}(s)}{\partial s} = \hat{\mathbf{u}}(s)\,\mathbf{R}(s), \\ \frac{\partial \mathbf{n}(s)}{\partial s} &= -\mathbf{f}(s), \quad \frac{\partial \mathbf{m}(s)}{\partial s} = -\mathbf{l}(s) \;-\; \mathbf{v}(s)\times\mathbf{n}(s). \end{aligned} \tag{2}$$

$\mathbf{r}(s)$ and $\mathbf{R}(s)$ describe the rod's spatial configuration and the orientation of its cross-sections, $\mathbf{v}(s)$ and $\mathbf{u}(s)$ represent the corresponding strains or geometric deformation measures, $\mathbf{n}(s)$ and $\mathbf{m}(s)$ represent the internal forces and moments generated to maintain equilibrium within the rod, $\mathbf{f}(s)$ and $\mathbf{l}(s)$ represent the distributed external loads acting on the rod. Thereafter, both static and kinematic models are developed, and numerical methods are applied to solve these models. The resulting motion and deformation outcomes of the designed RigFlex parallel robot are then rigorously evaluated. If the robot demonstrates favorable kinematic performance, the design is deemed successful; otherwise, the performance of the rigid parallel robot is reexamined for further improvement. Consequently, it is evident that the design process described above follows an iterative methodology.

III. Illustrative Examples of the 3-RRR Robots

In this section, we take the 3-RRR parallel robot as an example and apply the proposed design method to develop three types of RigFlex parallel robots.

A. Design of RigFlex Parallel Robots



(a) Schematic diagram of robot. (b) Photograph of the robot [23].

Fig. 2: 3-RRR parallel robot.

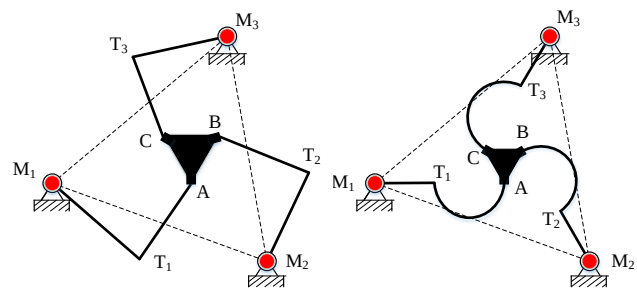
As shown in the Fig.2, the 3-RRR parallel robot is a type of parallel manipulator with three rotational DOFs, characterized by three rotating limbs connected to a fixed base, providing high precision and stiffness for tasks such as positioning and assembly in various industrial applications [1]. To simplify the analysis, the moving platform is designed as an equilateral triangle with side lengths AB = BC = CA = W.

Using the Grübler-Kutzbach (G-K) formula [24], we analyze the DOFs of the 3-RRR parallel robot:

$$\mathrm{DOF} = m(N-J-1)+\sum_{i=1}^{J} f_i = 3\times(8-9-1)+9\times 1 = 3 \tag{3}$$

where m is the system order; N is the number of links; J is the number of joints; f_i is the DOF of the i-th joint.

From Eq. (3), the DOFs of the 3-RRR parallel robot in its workspace are determined to be three. By introducing constraints, the robot can be transformed into either a zero-DOF or a negative-DOF robot. Based on the DOFs, the design is divided into two cases: $\mathrm{DOF} < 0$ and $\mathrm{DOF} = 0$.



(a) Rigid parallel robot (b) RigFlex parallel robot.

Fig. 3: Design of RigFlex parallel robot with $\mathrm{DOF} < 0$.

1) $\mathrm{DOF} < 0$: As shown in the Fig. 3a, after removing the two rotational joints from each leg of the 3-RRR parallel robot, the DOFs of the rigid parallel robot are given by:

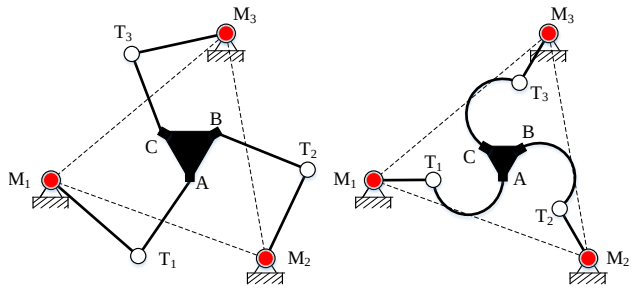
$$\mathrm{DOF} = m(N-J-1)+\sum_{i=1}^{J} f_i = 3\times(2-3-1)+3\times 1 = -3 \tag{4}$$

The DOFs of the rigid parallel robot become negative. By fixing one end of three flexible beams at points A, B, and C on the moving platform and the other end to the rigid links, a RigFlex parallel robot is formed, as shown in the Fig. 3b.

2) $\mathrm{DOF} = 0$: As shown in the Fig. 4a, after removing the one rotational joint from each leg of the 3-RRR parallel robot, the DOFs of the rigid parallel robot are given by:

$$\mathrm{DOF} = m(N-J-1)+\sum_{i=1}^{J} f_i = 3\times(5-6-1)+6\times 1 = 0 \tag{5}$$

Consequently, the DOFs of the parallel robot become zero. By rigidly attaching flexible beams at points A, B, and C on the moving platform and hinge-connecting their opposite ends to a rigid link, a RigFlex parallel robot is created, as shown in the Fig. 4b.



(a) Rigid parallel robot (b) RigFlex parallel robot.

Fig. 4: Design of RigFlex parallel robot with DOF = 0.

B. Kinetostatics Model of the RigFlex Parallel Robots

After completing the design of the RigFlex parallel robot, a kinetostatics modeling approach is employed to study its motion and deformation [25]. In the modeling process, it is assumed that the flexible beams are inextensible, and the influence of shear deformation is neglected, thus allowing the application of geometrically nonlinear Euler-Bernoulli beam theory for modeling the flexible beams [26].

Based on the robot's DOFs and actuation methods, three cases of the RigFlex parallel robot are modeled and discussed. The kinetostatics model of the RigFlex parallel robot consists of three components: constitutive equations, force equilibrium equations, and geometric compatibility equations [27].

1) The rest state of RigFlex parallel robots: The geometrical characterization of the rest state RigFlex parallel robot is identical for all three cases. As shown in Fig. 5 to Fig. 7, at its rest state where the flexible beams of RigFlex parallel robots are not deformed, we have

$$\begin{aligned} &x_A = 0;\ y_A = -d;\ x_B = d\cos\frac{\pi}{6};\ y_B = d\sin\frac{\pi}{6};\\ &x_C = d\cos\frac{5\pi}{6};\ y_C = d\sin\frac{5\pi}{6}; \alpha_1 = \alpha_2 = \alpha_3 = 0;\\ &d = \frac{W}{2\cos\frac{\pi}{6}}; \varphi_1 = 0;\ \varphi_2 = \frac{\pi}{6} + \frac{\pi}{2};\ \varphi_3 = \frac{5\pi}{6} + \frac{\pi}{2} \end{aligned} \tag{6}$$

Accordingly, under the rest state of the studied the RigFlex parallel robot the coordinates of the base joints M_1, M_2, and M_3 can be obtained:

$$x_{mi} = x_{0i} - l\cos\varphi_i;\ \ y_{mi} = y_{0i} - l\sin\varphi_i \tag{7}$$

where l is the length of rigid link.

2) Kinetostatics model: As shown in Fig. 5 to Fig. 7, based on the geometrically nonlinear Euler–Bernoulli beam theory, the constitutive equations for the three cases are derived as follows [28]:

$$\begin{aligned} &\text{D.E.} \quad \frac{\mathrm{d}^2\theta_i}{\mathrm{d}s^2} = -\frac{F_{yi}}{EI}\cos\theta_i(s) - \frac{F_{xi}}{EI}\sin\theta_i(s)\\ &\text{B.C.} \quad \theta_i(0) = \theta_{oi}\\ &\qquad \frac{\mathrm{d}\theta_i}{\mathrm{d}s}(L) = \frac{M_{oi}}{EI} + k_0(L) = \frac{M_{oi}}{EI} + \frac{1}{R} \end{aligned} \tag{8}$$

with

$$\begin{aligned} &x_i(s) = x_{Ti} + \int_0^s \cos\big(\theta_i(\xi)\big)\,\mathrm{d}\xi\\ &y_i(s) = y_{Ti} + \int_0^s \sin\big(\theta_i(\xi)\big)\,\mathrm{d}\xi\\ &s \in [0, L];\ \ i = 1, 2, 3. \end{aligned} \tag{9}$$

Using the vector method and static force analysis for rigid robots, the force equilibrium equations for the three cases are derived as follows:

$$\begin{aligned} &\sum_{i=1}^{3} F_i = 0;\ \sum_{i=1}^{3} F_{xi} = 0\\ &\sum_{i=1}^{3} M_{oi} + F_{x2}\,W\,\sin(\Phi) + F_{y2}\,W\,\cos(\Phi)\\ &-\ F_{x1}\,W\,\cos\Big(\tfrac{\pi}{3} - \Phi\Big) -\ F_{y1}\,W\,\sin\Big(\tfrac{\pi}{3} - \Phi\Big) = 0 \end{aligned} \tag{10}$$

Using the vector method, the geometrical compatibility relationships for the three cases are given as follows [29]:

$$\begin{aligned} &\Phi = \theta_1(L) - \frac{\pi}{2} = \theta_2(L) - \frac{7\pi}{6} = \theta_3(L) - \frac{11\pi}{6}\\ &|x_i(L) - x_{i-1}(L)| = |x_{ki} - x_{k(i-1)}|\\ &|y_i(L) - y_{i-1}(L)| = |y_{ki} - y_{k(i-1)}|; i = 2, 3\\ &|x_3(L) - x_1(L)| = |x_{k3} - x_{k1)}|\\ &|y_3(L) - y_1(L)| = |y_{k3} - y_{k1)}| \end{aligned} \tag{11}$$

- Case A

As shown in the Fig. 5, the RigFlex parallel robot is characterized by three revolute joints, with motor drives applied at the base joints M_1, M_2, and M_3.

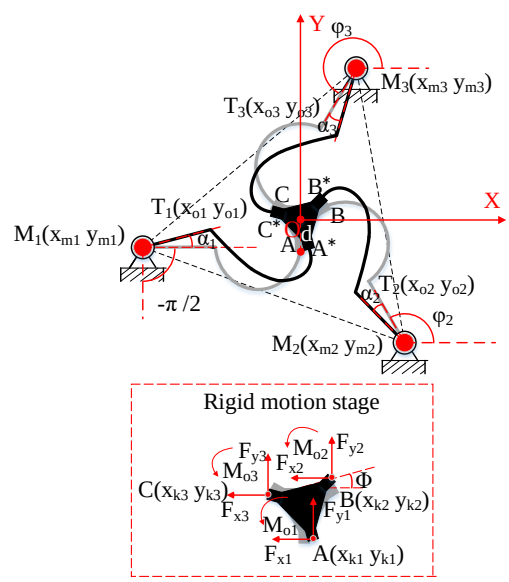


Fig. 5: RigFlex parallel robot of case A.

The constitutive relationships for this case are Eq. (8) and Eq. (9). The force equilibrium for this case is Eq. (10).

The geometrical compatibility relationships for this case are obtained by adding Eq. (11), Eq. (12) and Eq. (13).

$$\theta_{oi} = \alpha_i - \frac{\pi}{2} + \frac{2\pi}{3}(i-1) \tag{12}$$

$$\begin{aligned} x_{Ti} &= x_{mi} + l\cos(\alpha_i + \varphi_i) \\ y_{Ti} &= y_{mi} + l\sin(\alpha_i + \varphi_i) \end{aligned} \tag{13}$$

- Case B

As shown in the Fig. 6, the RigFlex parallel robot is characterized by six revolute joints. Motor drives are applied at the base joints M_1, M_2, and M_3.

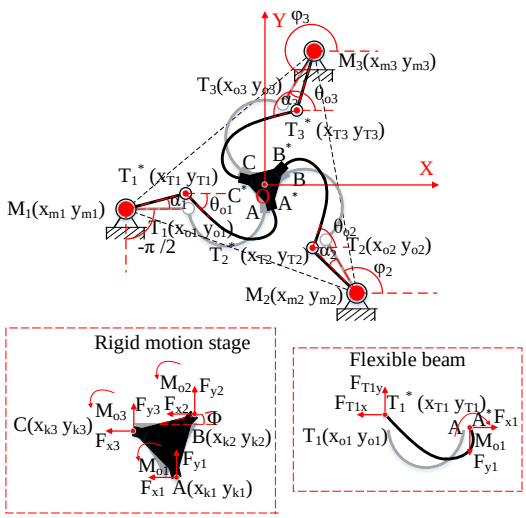


Fig. 6: RigFlex parallel robot of case B.

The constitutive relationships for this case are Eq. (8) and Eq. (9). The force equilibrium for this case is obtained by adding Eq. (10) and Eq. (14). The geometrical compatibility relationships for this case are obtained by adding Eq. (11) and Eq. (13).

$$\begin{aligned} &F_{Tix} = F_{xi};\ F_{Tiy} = F_{yi} \\ &M_{oi} + F_{Tiy}(x_i(L) - x_{Ti}) + F_{Tix}(y_i(L) - y_{Ti}) = 0 \end{aligned} \tag{14}$$

- Case C

As shown in the Fig. 7, the RigFlex parallel robot is characterized by six revolute joints. Motor drives are applied at the joints T_1, T_2, and T_3.

The constitutive relationships for this case are Eq. (8) and Eq. (9). The force equilibrium for this case is obtained by adding Eq. (10), Eq. (14) and Eq. (15). The geometrical compatibility relationships for this case are obtained by adding Eq. (11), Eq. (12) and Eq. (16).

$$\begin{aligned} &F_{Tix} = F_{mix}; F_{Tiy} = F_{miy} \\ &M_i - M_{oi} - F_{Tiy}(x_i(L) - x_{Ti}) - F_{Tix}(y_i(L) - y_{Ti}) = 0 \\ &M_i + F_{Tiy}(x_{Ti} - x_{mi}) + F_{Tix}(y_{Ti} - y_{mi}) = 0 \end{aligned} \tag{15}$$

$$\begin{aligned} x_{Ti} &= x_{mi} + l\cos(\alpha_i - \varphi_i) \\ y_{Ti} &= y_{mi} + l\sin(\alpha_i - \varphi_i) \end{aligned} \tag{16}$$

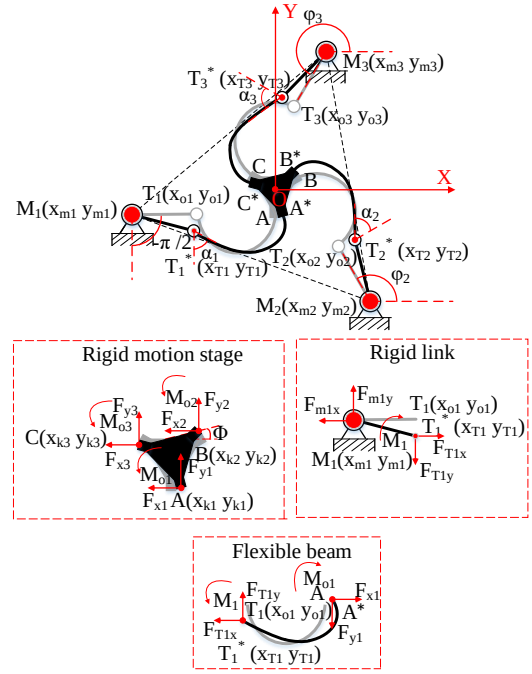


Fig. 7: RigFlex parallel robot of case C.

C. Numerical Results and Analysis

After completing the kinetostatics models of the RigFlex parallel robot, numerical methods were employed to compute the models for the three cases. Nonlinear equations or systems of equations were solved using the Newton-Raphson method, while the BVP was addressed using the Galerkin method [30].

1) Parameter setup: The following provides the material and geometric details:

$$E = 200 \times 10^9\,\text{Pa}; \nu = 0.30; w = 0.01\,\text{m}; h = 0.004\,\text{m};$$

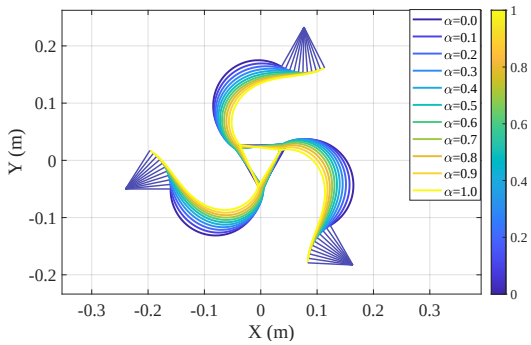
$$R = 0.08\,\text{m}; W = 0.05\sqrt{3}\,\text{m} L = \pi R; I = \frac{h^3 w}{12};$$

$$G = \frac{E}{2(1+\nu)}; A = wh; d = \frac{W}{2\cos\frac{\pi}{6}}; l = 0.08\,\text{m}$$

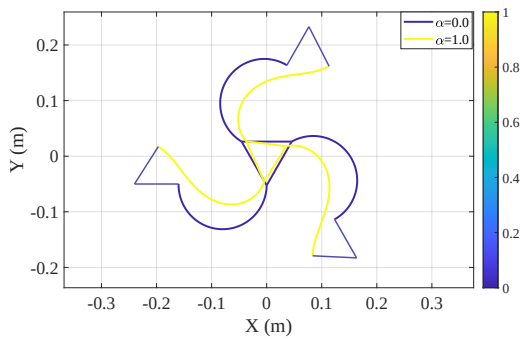
In this numerical test, only the rotational angle of the kinematic joint is controlled:

$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha = \left[0.1, 0.2, 0.3, 0.4, \ldots, 1\right]^{\mathrm{T}}$$

2) Analysis and discussion: The motion of the RigFlex parallel robot in case A, with motor drives applied to base joints M_1, M_2, and M_3, is shown in the Fig. 8. The motion of the RigFlex parallel robot in case B, with motor drives applied to base joints M_1, M_2, and M_3, is shown in the Fig. 9. The motion of the RigFlex parallel robot in case C, with motor drives applied to joints T_1, T_2, and T_3, is shown in the Fig. 10. It can be observed that the RigFlex parallel robot is capable of reaching different positions under varying motor angle controls, achieving precise position control. To provide a clearer visualization of the deformation of the flexible beams during the motion of the designed RigFlex parallel robot, a comparison is made between the robot's position in its rest state and its position in its final state,

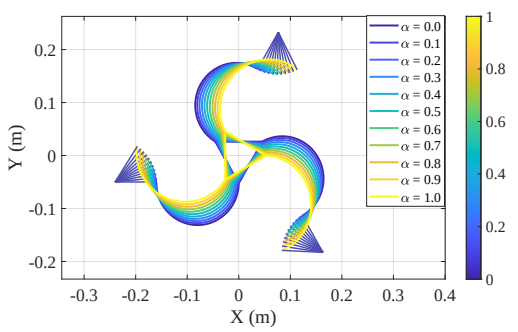


(a) Overall motion.

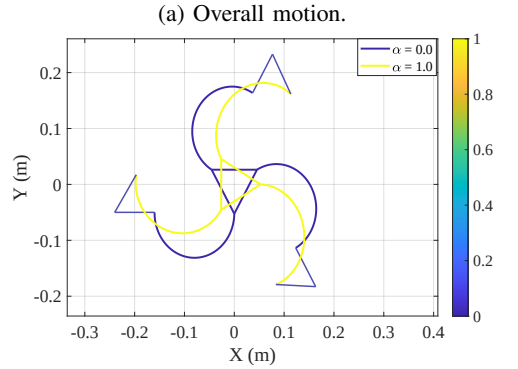


(b) Rest state vs final state.

Fig. 8: Numerical result of case A.

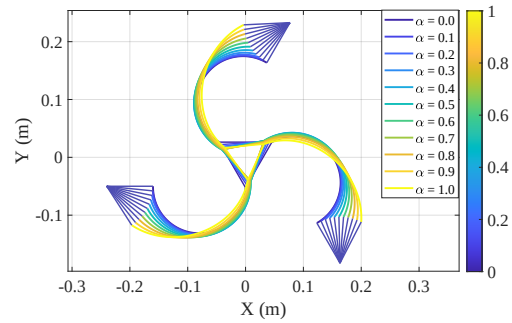


(a) Overall motion.

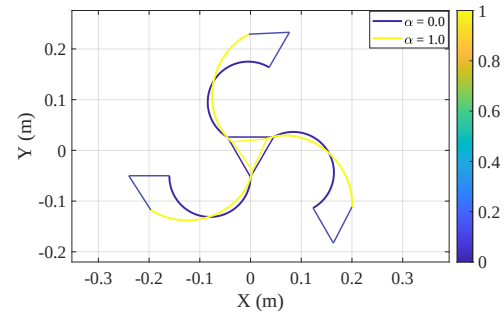


(b) Rest state vs final state.

Fig. 9: Numerical result of case B.



(a) Overall motion.



(b) Rest state vs final state.

Fig. 10: Numerical result of case C.

as shown in the Fig. 8b, Fig. 9b, and Fig. 10b. It can be observed that, in each case, the three flexible beams of the designed RigFlex parallel undergo significant deformation during motion, indicating that the designed RigFlex parallel robot operates through the combined action of the rigid motion of the rigid links and the deformation of the flexible beams. The robot achieves precise position control while maintaining the compliance characteristic of flexible robots, thus integrating the advantages of both rigid and flexible robots. These results demonstrate the feasibility of the design methodology proposed in this paper. To ensure that the flexible beams in the RigFlex parallel robot undergo deformation, this paper only discusses the cases where the DOFs are greater than or equal to zero. However, in constraint systems with DOFs greater than zero but less than the maximum achievable within the workspace, the application of excessive drive may nevertheless induce deformation of the links.

IV. CONCLUSIONS

Existing research on RigFlex parallel robots lacks a specific design methodology. In this context, this paper analyzes the DOFs of rigid robots and the relationship between their constraints and actuators, leading to the development of a design methodology for RigFlex parallel robots based on DOFs. Using the proposed method, three types of RigFlex parallel robots were designed, with the 3-RRR parallel robot serving as a case study. The kinetostatics models of the three designed RigFlex parallel robots were developed, and the models were solved using numerical methods to obtain the motion and deformation results of the RigFlex robots. The results demonstrate that the designed RigFlex parallel

robots combines the precise control accuracy of a rigid robot with the excellent compliance of a flexible robot, thereby confirming the feasibility of the design methodology. The proposed design methodology addresses a critical gap in the field of RigFlex robotics, providing a systematic framework to guide the design process of such robotic systems. Future research may focus on a comprehensive analysis of the workspace and singularity characteristics of the designed RigFlex parallel robots, as well as an in-depth investigation into the deformation behavior of alternative beam configurations. Additionally, further studies could explore the design methodology of RigFlex parallel robots under over-actuated conditions, wherein the DOFs of the parallel mechanism are constrained to be less than the maximum theoretical DOFs within the workspace but remain greater than zero. These efforts will contribute to a more complete validation and refinement of the proposed design methodology for RigFlex parallel robots.

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